Statistics
Lecture 07
Hypothesis testing

Case: Test for μ in normal distribution with known variance (MATLAB: ztest)

- Model
- Obs
- Hypothesis: 2-sided/1-sided
- Purpose of test
- Estimator for μ and its distribution
- Test statistic Z_0
- A = Accept region, C = Critical (reject) region
- Error types 1 and 2
- Level of significance, α = maximal acceptable probability of type 1 error
- α → A,C 2-sided/1-sided (MATLAB: norminv)
- Small α and big α
- Type 2 error, β = β(μ)
- α vs. β
- Power, γ = γ(μ) = 1 - β(μ)
- p-value: definition, 2-sided/1-sided (MATLAB: normcdf)

Case: Test for μ in normal distribution with unknown variance (MATLAB: ttest)

- Estimator for variance
- Test statistic T_0
- α → A,C 2-sided/1-sided (MATLAB: tinv)
- p-value (MATLAB: tcdf)

Case: Test for σ^2 in normal distribution (MATLAB: vartest)

- Hypothesis
- Estimator for variance
- Test statistic, χ_0^2
- Critical region and p-value (MATLAB: chi2inv, chi2cdf)